

MU-HUA (MAURICE) WANG

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RESEARCH INTEREST

Physical assistive robotics for long-term care. I want to build robots that perform contact-rich care tasks on a bedridden body, starting with diaper changing. The task demands three things at once: perceiving a body that will not cooperate, knowing enough about that person to move them without harm, and handling deformable material safely against skin. My work covers the first two and has begun the third: a clinically validated markerless motion capture system, studies of what a robot must know about a body, and a diaper-changing simulation platform in progress.

EDUCATION

National Yang Ming Chiao Tung University — M.S., Biomedical Engineering

Entered Feb 2025 · on approved leave of absence since May 2025, enrollment retained · researching full time

National Yang Ming Chiao Tung University — B.S., Biomedical Engineering · Sep 2020 – Jun 2024

– GPA 3.72 / 4.30 · final two years 4.07 / 4.30 · **IELTS 8.0** (R 9.0, L 8.5, S 7.5, W 6.5)

RESEARCH EXPERIENCE

Project pages, videos and full manuscripts for everything below: maurice1128.github.io

Care-Robot Simulation Platform — independent · 2026 – present · work in progress

Measuring Whether a Care Robot Hurts the Patient: robot-assisted diaper changing and perineal cleaning

- A MuJoCo platform being built to measure three things separately on one statistical (SMPL-X) body: garment state, cleaning coverage and deep-tissue strain. None of the three is complete; every result is simulated and model-internal, not validated against measurements on people.
- Checked so far: a sacral soft-tissue model whose energy passes finite-difference and patch tests ($6-7 \times 10^{-12}$); on a rigid bed over 285 simulated minutes, skin pressure falls while deep compressive strain rises, the two strain summaries agreeing within 1% between the two finest meshes; side-lying, a 10 mm tool's geometric access to the perineal target rises from 19% to 83% when the upper leg is flexed 45° and abducted 30° . Video on the website.

Robot Learning and Embodied AI — independent · Oct 2025 – present

Sole author on five manuscripts, **two submitted**. Each result is tested against controls built to break it, and its limitations are stated.

Gentle by knowing: person knowledge, not force thresholds, protects restricted joints in simulated robotic arm lifting — *submitted, Frontiers in Robotics and AI (Brief Research Report)*

- Musculoskeletal simulation of a robot lifting the forearm of simulated persons with restricted shoulders or elbows; controllers differ only in what they know about the person and whether they may decline. **No force-magnitude guard cut harmful outcomes by significantly more than the 5 percentage points fixed in advance:** 95–100% of harmful outcomes occur in the first attempt. From a record with 20° error, a policy learned offline in simulation cut them from 110 to 25 of 300 by declining 91 (174 vs 177 safe completions).

Withdrawn Teachers and Learned Body Models in Muscle-Driven Motor Learning — *submitted, IEEE TCDS*

- One SAC learner on six simulated muscle-driven bodies (MyoSuite elbow and finger, planar arms of one to four joints); a withdrawn teacher against a frozen copy of the student (same imitation term, no teacher), a babble-trained body model against a random one. On the finger the withdrawn learner ends 43 mm ahead of one never guided and keeps more the longer it was taught (24% of the teacher's worth at 2,000 steps, all of it at 8,000); on the arms a random model harms more with every joint added, the trained one does not.

Learned In-Betweeners Make Simulated Dexterous Hands Learn Faster — *manuscript, prepared for IEEE RA-L*

- Tested each layer of a modular motion model on simulated hands, main comparisons declared before the runs. Keyframes joined by a learned in-beweeners lead a continuous latent at the same decision rate on held-out skill sequences (0.59 vs 0.22 success at 1.5 M frames, 17 of 20 seeds); at 5 M the data do not separate them. Picking the best policy per stage completes 97 of 400 test episodes vs 61; separately trained skills, 45.

Partial labels hide compositional difficulty in hand-motion data — *manuscript, prepared for TMLR*

- Calibrated a paired held-out-combination penalty on four public hand-motion datasets against zero-truth (+2.6%), planted and permuted-grid controls. On OakInk-Image, a correct label covering only the first half of each

trajectory, at matched data volume, cut the penalty from +17.0% to +3.9%, at most 2.9 points above the control; a TACO penalty written down before its data were opened was found (+10.6%).

In draft: *Reparameterisation Invariance Limits the Benefit of Reorganising a Body Code for Morphology Transfer*, where an algebraic identity rules out a family of proposals before any experiment runs.

RGB Multi-View 3D Markerless Motion Capture — NYCU · Oct 2023 – Sep 2025

Research Assistant, then Team Lead · Advisor: Prof. Hui-Ting Shih · the perception layer a robot over a bed needs

- **Built the real-time pipeline end to end:** GPU acceleration and CPU parallelism across capture, inference, triangulation and inverse kinematics; DLT with occlusion handling; and **ROWV**, an outlier-rejection algorithm beating KNN and RANSAC on this problem. **Validated** against VICON over 21 participants, plus 104 walking trials on a nine-camera ring.
- **Asked how much sensing error the downstream decision can absorb:** in forward-dynamic simulation, identification of a weak muscle group falls $0.845 \rightarrow 0.549$ as joint-angle error grows $0^\circ \rightarrow 12^\circ$.
- **Ran the collaborations and the field work:** 2 hospital MOUs (Taipei Veterans General; Taipei City Hospital), 200+ stroke and knee-pain patients in Vietnam, 40+ at Kaohsiung Chang Gung, 20+ elderly at a daycare centre, amputee trials with NYCU Physical Therapy.
- **Shipped the product** and led a 5-person team: PyQt6 clinician GUI, SQL and cloud pipeline with ICD-10 coding, Unreal Engine rehabilitation game; 6 media features.

Behavioural Trajectory Modelling — Taoyuan General Hospital · Jan 2023 – present

Advisor: Dr. Tso-Jeng Wang. Mapped methadone dose trajectories to a polar angle, then clustered with a bivariate von Mises mixture on the torus. 9,900 urine tests from 552 patients.

EEG and EMG Sleep–Epilepsy Analysis — NYCU · Oct 2022 – Oct 2023

Advisor: Prof. You-Yin Chen. Built a pipeline processing 24-hour recordings in 3 minutes with automatic WAKE / NREM / REM staging; designed a surgical modification that raised EMG signal-to-noise.

PUBLICATIONS

Peer-reviewed conference papers

1. **Wang, M.-H.**, et al. *Markerless Gait Analysis for Lower Back Pain*. SEMBA 2025. **(first author)**
2. Huang, L.-W., **Wang, M.-H.**, et al. *A Real-Time Algorithm for Detecting Gait Events in Lower-Limb Prosthesis Users*. ASB 2025.
3. Huang, A.-Y., **Wang, M.-H.**, et al. *Spatiotemporal Cortical-Synergy Connectivity in the Gait Cycle*. SEMBA 2025.
4. Chang, C.-W., **Wang, M.-H.**, et al. *Central Thalamic Deep Brain Stimulation and Sleep in an Alzheimer's Model*. ISMRM 2024.
5. Chen, B.-H., et al. incl. **Wang, M.-H.** *Robust Self-Occlusion Mitigation in Multi-View 3D Markerless Motion Capture*. ICMST 2024.

Submitted (sole author, 2026). *Frontiers in Robotics and AI* · IEEE TCDS. **Manuscripts** prepared for IEEE RA-L and TMLR.

Drafts (on the website). One in robot learning, three in clinical biomechanics, one in behavioural data science.

TEACHING

Physical Science, Grade 8 — Tamkang High School · Feb – Aug 2026

Class 8 rose to 2nd of eight classes, its margin over the school average widening $+5.1 \rightarrow +11.8$ points. **The gain came from the bottom:** its bottom-12% cutoff finished 28 points above the school's, up from 14.

SKILLS AND AWARDS

ML / Robotics — Python, MATLAB, C/C++ · PyTorch · SAC, PPO, model-based RL, offline policy learning · MuJoCo, MyoSuite, Shadow Hand, SMPL-X

Systems / Statistics — CuPy, Numba, PyQt6, SQL · paired designs, permuted-label controls

Awards — 2nd Place, Youth AI Startup Hackathon, Ministry of Digital Affairs (2023) · 1st Place, Elderly Health & Technology, NCNU (2023)